ML Security and Abuse Pathways Audit Report

This report evaluates the security and privacy vulnerabilities of a recidivism prediction model built on the ProPublica COMPAS dataset. The analysis focuses on three key risk areas: PGD evasion attacks, label-flip data poisoning and membership inference (MI). These risks are especially critical in criminal justice applications, where model failures can reinforce systemic bias and lead to harmful real-world consequences.

PGD Evasion Attack Audit

The objective of this analysis was to assess model robustness by applying Projected Gradient Descent (PGD) perturbations. Both Logistic Regression (LR) and Gradient Boosted Trees (GBT) were found to be vulnerable, as even small perturbations ($\varepsilon = 0.25$) significantly increased False Positive Rates (FPR) across racial groups.

From a fairness perspective, the Adverse Impact Ratio (AIR) for LR decreased from 1.961 to 1.535. However, this apparent improvement is misleading, as it is driven by widespread misclassification rather than genuine fairness gains. At higher perturbation levels, model performance deteriorates further, reflecting instability.

Comparatively, LR is more sensitive to adversarial noise than GBT, showing faster fairness degradation. These findings demonstrate that robustness failures directly translate into fairness risks.

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--- PGD Attack Results for Logistic Regression ---
 epsilon  FPR_AA  FPR_CA    AIR  delta_AA  delta_CA
  0.2500  0.5690  0.3700 1.5350    0.2880    0.2270
  0.5000  0.7910  0.5600 1.4110    0.5100    0.4170
  1.0000  0.9780  0.8840 1.1060    0.6970    0.7410
  2.0000  1.0000  1.0000 1.0000    0.7190    0.8570
AIR does not fall below 0.80 within the tested epsilon range.

Running PGD Evasion Audit for Gradient-Boosted Tree (GBT) using a

--- PGD Attack Results for Gradient-Boosted Tree (with LR surrogat
 epsilon  FPR_AA  FPR_CA    AIR  delta_AA  delta_CA
  0.2500  0.5530  0.3510 1.5770    0.2360    0.1730
  0.5000  0.7330  0.5140 1.4260    0.4160    0.3360
  1.0000  0.8990  0.7600 1.1820    0.5820    0.5830
  2.0000  1.0000  0.9900 1.0100    0.6830    0.8120
AIR does not fall below 0.80 within the tested epsilon range.
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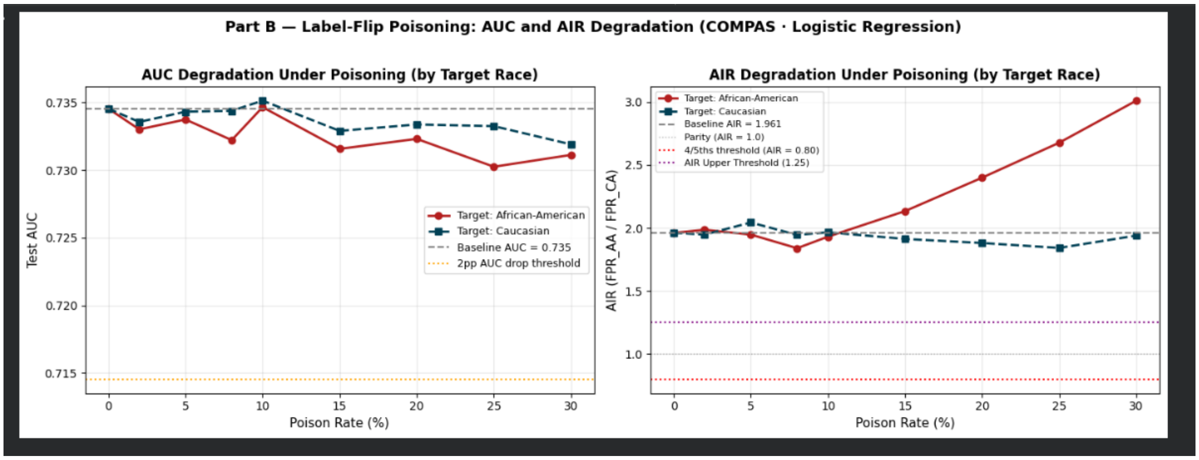
Poisoning Loop with Fairness Monitoring (Highest Risk)

This section evaluates the impact of targeted label-flip poisoning identified as the most critical risk. When high-risk labels were flipped, AIR on African American poisoning sweep increased significantly from 1.961 to 2.399 at 20% poisoning and reached 3.010 at 30%, indicating a severe amplification of racial disparities.

The attack remains highly stealthy. Model performance metrics such as AUC decreased steadily (0.735 to 0.732), while PSI values stayed near zero, meaning traditional monitoring systems failed to detect the issue.

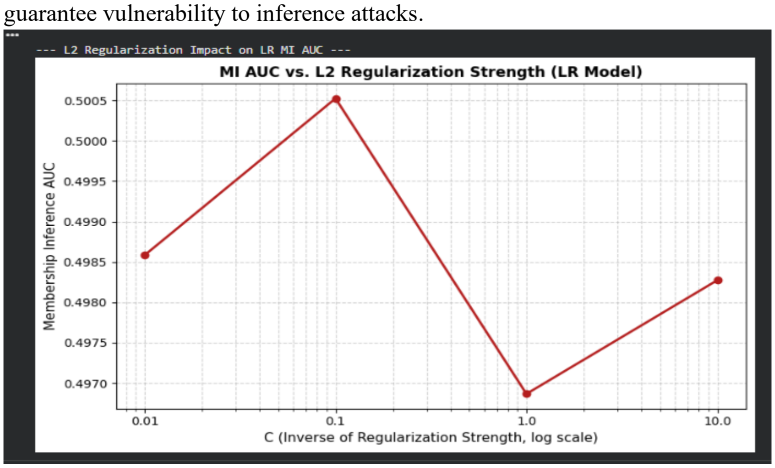
Additionally, poisoning any subgroup worsens disparities, confirming that the model amplifies existing bias structures. This highlights a dangerous failure mode where fairness collapses without visible performance degradation.



Membership Inference (Privacy Risk)

Membership inference analysis was used to evaluate whether the model leaks information about its training data. It showed minimal privacy risk. MI AUC scores were approximately 0.497 for LR and 0.500 for GBT, equivalent to random guessing. Confidence distributions between training and test data overlapped significantly.

Although GBT showed a higher generalization gap, this did not increase privacy risk, suggesting that overfitting alone does not guarantee vulnerability to inference attacks.



Discussion

This audit highlights that robustness, fairness, and security are deeply interconnected. Adversarial attacks can significantly degrade fairness without affecting traditional performance metrics, while standard monitoring tools fail to detect these risks. This underscores the need for fairness-aware evaluation in high-stakes models.

Mitigation Strategies

A proactive mitigation strategy involves subgroup-specific data monitoring, tracking label distributions to detect abnormal shifts and preventing poisoned data from entering the model.

A reactive strategy is fairness-aware threshold adjustment, which can reduce disparities (for example, lowering AIR toward 1.5). However, may increase false negatives, creating a tradeoff between fairness and overall accuracy.

Conclusion

The most critical finding is the model's vulnerability to stealthy label-flip poisoning, which significantly increases racial disparities without detection. While privacy risks are low, fairness and security risks are substantial.

Responsible deployment requires continuous fairness monitoring, robust data validation, and careful model selection to ensure systems are not only accurate, but also fair and secure.